



Leveraging explainable AI for sustainable agriculture: a comprehensive review of recent advances

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Abstract

In recent years, Artificial Intelligence (AI) has emerged as a disruptive force, transforming industries worldwide through its ability to analyze vast amounts of data and generate informed decisions. The agricultural sector is undergoing a revolutionary transformation driven by the adoption of this technology to enhance productivity, efficiency, and sustainability. However, the lack of transparency in AI models has sparked doubts about their dependability, particularly when applied to critical industries like medicine and agriculture. To address these challenges, research into Explainable Artificial Intelligence (XAI) gained momentum in early 2017. Tools like Local Interpretable Model-agnostic Explanations (LIME), SHapley Additive Explanations (SHAP), and Gradient-weighted Class Activation Mapping (Grad-CAM) helped make progress. These methodologies have proven instrumental in elucidating the internal workings of complex AI models, making their decision-making processes more transparent and comprehensible. This review surveyed current agricultural developments, including using AI to identify plant diseases. We explored articles from Scopus, Web of Science, IEEE Xplore, PubMed, Google Scholar, Agricola, and the Association for Computing Machinery, published between 2017 and 2024. After extracting and filtering, we included publications that satisfied the selection criteria in the review. It has provided an overview of the uses of XAI methods and the difficulties researchers have encountered. In addition, we reviewed recent advancements in machine learning (ML) and deep learning (DL) to find the limitations that agriculturists face in agriculture. We also provide a modern explainable model for identifying plant diseases and show how XAI can be used in agricultural applications. The conclusion of this paper indicates how XAI is necessary for ML, DL models as well as real time monitoring in the agriculture field.

Keywords Machine learning · Deep learning · Explainable artificial intelligence · Smart agriculture · Disease recognition

1 Introduction and scope

Artificial Intelligence (AI) is the term used to describe the development of computer systems that can do any work by thinking and acting like humans. AI allows computer systems to learn from training data to solve particular problems and perform functions in the real world by imitating human intelligence (Saranya and Subhashini 2023). The motivation for constructing AI was not just to replicate human performance but occasionally to outperform it (Hagras 2018). Alphabet Inc.'s current CEO believes AI will potentially have a greater impact on humanity than fire, electricity, and the internet (Knowles 2021). Besides, AI algorithms enable systems to make decisions based on input data. AI encompasses various techniques, models, and approaches to analyze data, recognize patterns, make choices, and act, including machine learning (ML), deep learning, natural language processing, computer vision, robots, and more. These algorithms and cutting-edge research fields produce highly accurate results and predictions from the input data. According to a statistic, almost 70% of businesses desire to include AI into their operations by 2030, indicating a significant increase in the use and acceptance of AI (Bughin et al. 2018). Nevertheless, the use of AI and ML technologies beats traditional models, such as generalized linear models, in predicting accuracy. This change is mainly caused by methods like random forests and neural networks (Breiman 2003), which get around the usual steps needed to process data and effectively figure out complicated data patterns (Ryo and Rillig 2017).

On the other hand, numerous industries, including education, healthcare, transportation, agriculture, business, insurance, and even the military, use these AI models in various settings. Schools use AI to enhance students' learning and assessments (Pham and Shen 2017), and hospitals use it to help doctors diagnose and identify illnesses quickly and effectively. Banks use AI models to make decisions about loans and investments (Pham and Shen 2017; Rachha and Seyam 2023; Chong et al. 2017), and vehicles use them to prevent accidents. AI technology has indeed permeated many aspects of our daily lives. Diverse fields, from education and healthcare to finance and transportation, witness its influence. Businesses also capitalize on AI models to streamline operations and improve customer experiences. Moreover, AI is advancing to revolutionize and enhance efficiencies across various agricultural industries to boost profitability while minimizing adversarial effects (Gardezi 2024; Liakos et al. 2018). According to the statistics from the web, the contribution of AI applications in agriculture to the global economy reached \$8.1 billion in 2023, and the worldwide market for AI in agriculture is projected to exceed \$15.3 billion by the year 2030. Again, autonomous farming systems driven by AI are projected to represent 40% of agricultural equipment sales by 2027 (Bhattacharya 2024). The statistical data demonstrate the transformative impact of AI on agrarian development, particularly in areas such as crop management, harvesting, pest control and prevention, yield prediction, weed control, and pond management.

1.1 Motivation and contribution

AI and XAI are becoming more significant in several agricultural industries, such as farm management and decision-making support (Nott 2017). Most autonomous systems using ML and DL models are "black boxes" that predict agricultural diseases without clarity, preventing confidence, adoption, and practical decision-making among farmers, agronomists, and policymakers. XAI helps people understand why and how AI models produce

predictions by making models interpretable and visible. In the case of agrarian yield prediction, XAI can assist the farmer in forecasting the yield by considering information about the soil, weather, and pests. Besides, XAI can clarify how the models recommend animal feed, breeding plans, or monitoring animal health regarding farm management. So, XAI and interpretable machine learning are getting more attention as ways to learn more about how AI makes decisions in critical areas like crop yield estimates and innovative agricultural systems (Rudin 2019; Rudin et al. 2022).

This study presents a comprehensive literature review on the agricultural revolution with XAI. In this study, most studies focusing on the application of AI and XAI in agriculture mainly highlight the significance of machine learning and deep learning. However, a distinct gap exists in the existing literature, as there are currently no studies that specifically address the application of XAI within the agricultural sector, including areas like crop yield prediction, plant disease detection, and animal disease identification. This survey offers valuable insights for professionals in agriculture regarding the applications of AI and XAI in the agricultural sector. By highlighting the areas, the study expects to foster the development of innovative solutions that will ultimately benefit farmers and the farm sector. We aim to motivate additional progress in this domain by investigating XAI's possibilities in agriculture. Hence, the contributions of this paper are given as follows:

- This paper provides XAI's fundamental concepts, terminology, and implementation challenges and opportunities across different sectors.
- The study's objective is to explore the influence of machine learning and deep learning models within the agricultural domain.
- The research thoroughly assesses the latest trends, challenges, and limitations in advancing XAI models for agriculture, offering future directions for inquiry.
- The study introduced a framework utilizing XAI to automate agricultural systems, focusing on disease recognition to enhance productivity and support sustainable farming practices.

1.2 Outline of the paper

The rest of the paper is outlined as follows: Sect. 2 presents the literature review method and search strategy. Section 3 provides a broad description of the idea of XAI for agriculture. Section 4 addresses the challenges and opportunities of XAI. We have discussed the application domains in Sect. 5. Section 6 presents a thorough assessment of the current level of developing research interest in agriculture by methodically analyzing and assessing the existing approaches while reviewing various papers on ML and DL-based models in agriculture. In Sect. 7, we present the survey discussion, including the study's findings, challenges, limitations, and research gaps of XAI's use in agriculture, and also provide future directions. Figure 1 shows the architecture of the overall organization of the study.

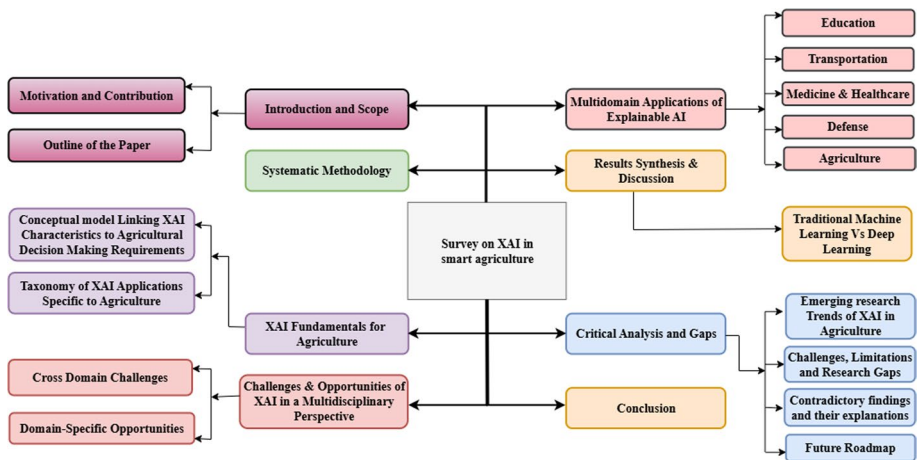


Fig. 1 Detailed representation of the survey's complete organizational framework, from design to analysis

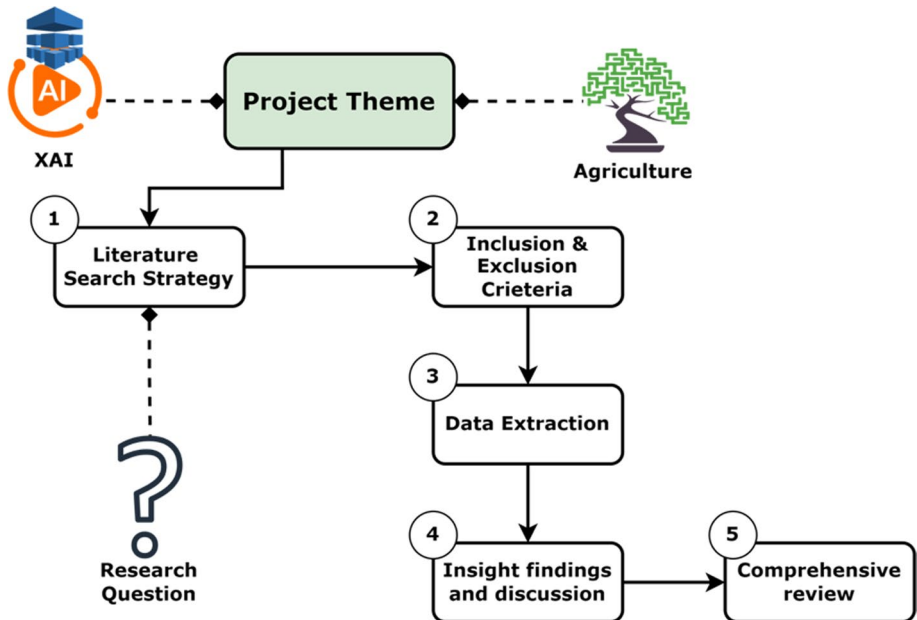


Fig. 2 Overall method for finding the literature and comprehensive reviews

2 Systematic methodology

This section provides the research question, method, and search strategy to extract the number of papers. Figure 2 shows the architecture of the overall process of the study to find the paper and extract the insights from the literature. This figure is classified into five significant steps. In the initial steps, we have searched the literature reviews based on the five research

questions. Then, we applied the inclusion and exclusion criteria to filter the literature review. Then, we used the data extraction and insight findings in the relevant discussion. Finally, we have performed comprehensive reviews of the adopted papers.

The study aims to address five research questions. (RQ1): How do different researchers conceptualize and define XAI in the literature, and what are the application domains, benefits, and challenges of XAI? (RQ2): What are the current research trends of ML, DL, and XAI in agriculture? (RQ3): Can XAI be deployed to identify plant disease? (RQ4): What XAI methods are most effective for real-time crop disease detection? (RQ5): How do different stakeholder groups (farmers vs. agronomists) interpret XAI explanations?

Based on the goal of this work and research questions, a literature search was conducted to complete the review. The flow diagram in Fig. 3 illustrates the schema for selecting relevant scholarly articles for our comprehensive review. The Systemic Literature Review adhered to the four-phase PRISMA process: identification (literature search), screening (title/abstract and full-text review), eligibility (detailed assessment), and inclusion (final

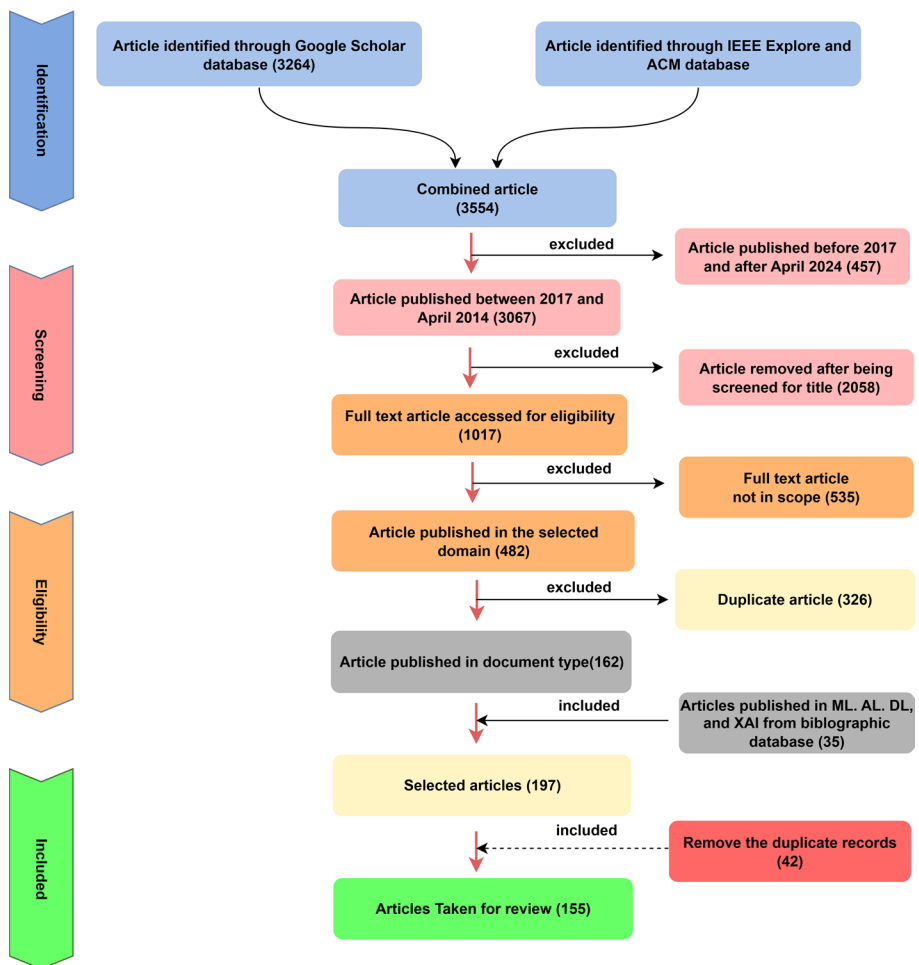


Fig. 3 Flow diagram of the search strategy and paper selection

synthesis). We concentrated on peer-reviewed studies from 2017 to April 2024 to identify “recent advances” in explainable AI (XAI) applications for sustainable agriculture, characterized as AI methodologies that enhance resource efficiency, safeguard the environment, and ensure long-term productivity. We did a thorough search of seven electronic databases: Scopus, Web of Science, IEEE Xplore, PubMed, Google Scholar, Agricola, ACM. These were chosen because they cover a wide range of fields (AI, agriculture, sustainability). The search was done using linguistic limits. The search phrases came from four main ideas: (1) explainable AI (XAI), (2) artificial intelligence/machine learning/deep learning, (3) agriculture, and (4) sustainability. Synonyms and a regulated vocabulary, such as MeSH phrases like “Explainable Artificial Intelligence” and “Precision Agriculture,” were used. To combine phrases, Boolean operators were used: OR for synonyms inside concepts and AND for concepts that overlap. Truncation () caught variations like “agricultur”, and phrase searching (“ ”) made sure that the search was specific (like “explainable AI”). The main search string was: (“explainable AI” OR “XAI” OR “interpretable AI” OR “transparent AI” OR “AI explainability”) AND (“artificial intelligence” OR “machine learning” OR “deep learning”) AND (agricultur* OR “precision farming” OR “smart agriculture” OR “crop management” OR “soil health” OR “irrigation”) AND (sustainab* OR “precision agriculture” OR “resource efficiency” OR “environmental impact” OR “climate resilience” OR “yield optimization”).

The eligibility criteria were predetermined according to PICOS (Population: agricultural systems; Intervention: XAI techniques; Comparator: non-XAI AI or traditional methods; Outcomes: sustainability metrics such as yield, resource use, or environmental impact; Study design: empirical or review studies) and were consistent with PRISMA item 6 (eligibility criteria). Two separate reviewers (Reviewer A: AI specialist; Reviewer B: agronomist) evaluated these at the title/abstract and full-text stages, with any inconsistencies addressed through conversation or by a third reviewer (senior author).

We did two independent reviews for screening three phases: title/abstract screening ($n = 3554$), full-text eligibility ($n = 354$, estimated), and data extraction ($n = 78$). Cohen’s kappa (k) statistic was used to quantify IRR since it looks at agreement that goes beyond chance. We understood k values to mean: < 0.00 (poor), $0.00–0.20$ (slight), $0.21–0.40$ (fair), $0.41–0.60$ (moderate), $0.61–0.80$ (considerable), and $0.81–1.00$ (nearly perfect). Percent agreement (%) was also supplied for transparency, since k might make agreement look lower than it really is in datasets that aren’t balanced. Two reviewers separately categorized each record as “Include,” “Exclude,” or “Uncertain” according to specified inclusion/exclusion criteria (e.g., XAI utilization, agricultural emphasis, sustainability results). Measures the level of agreement that goes above chance for two raters. Using a 2×2 contingency table (Include vs. Exclude, combining “Uncertain” with “Include” for binary analysis, as is common in SLR). Full texts checked for eligibility, such as how clear the XAI technique is and how well the sustainability measures work. Extracted data (e.g., XAI method, sustainability outcome) from the studies that were included.

For all of the studies that were included, we used techniques that were specific to the designs to check for quality (risk of bias): QUADAS-2 for diagnostic/prognostic studies (common in XAI for crop monitoring), ROBINS-I for non-randomized treatments, and AMSTAR-2 for reviews.

The identification phase yielded 3264 articles from Google Scholar, Scopus, PubMed, Agricola, Web of Science and 290 articles from IEEE Xplore and ACM databases for 3554

articles. Following the deduplication process, 487 articles were excluded as they were either published before 2017 or after April 2024, which did not meet the review's temporal criteria. This exclusion led to 3067 remaining articles. The screening phase further narrowed with 2050 articles excluded after title screening and an additional 535 articles excluded because the full texts were not within the survey's scope, leaving 1017 articles. During the eligibility phase, articles were subjected to more rigorous criteria—482 articles were excluded as they were not published in the selected domain, and 320 were duplicates, leaving 162 articles in the document type of interest. The inclusion phase focused on articles published in machine learning (ML), artificial intelligence (AI), deep learning (DL), and explainable AI (XAI) within the field of agriculture from the bibliographic database, which accounted for 35 articles. After adding 35 articles, we have found the total number of papers to be 197. Then, we have seen some duplicate records of 42 papers and discarded them to perform the final view. After final assessment, 155 articles were selected. The selected articles were thoroughly reviewed and included in the survey, providing a foundation for the analysis and discussion of various models used in the agricultural domain, emphasizing identifying the model with the highest accuracy as reported in the studies. This meticulous process, as depicted in the flow diagram, ensures a robust and exhaustive collection of literature pertinent to the intersection of AI technologies and agriculture.

3 XAI fundamentals for agriculture

The 'knowledge engineering phase' of the late 1970s and early 1980s focused on constructing expert systems that mimic human thinking in high-value domains like medical, engineering, and geology, leading to interest in AI system explainability (Ribeiro et al. 2016). We realized that explanations were crucial for system development, particularly for testing and end-user trust. In modern AI development, "interpretability" emphasizes explainability, highlighting the need for human-friendly machine-learned models (Preece et al. 2018). The concepts of interpretability and explainable AI might be misleading (Rudin 2019). Definition of "explanation": verification of AI system output judgments (Das and Rad 2020). In contrast, Poul Voosen defines interpretability as the technical issue of explaining AI algorithm results (Voosen 2017). The fundamental aim for creating XAI was to make AI-based systems more accurate, human-readable, intelligible, transparent, trustworthy, and fair. Figure 4 contrasts black, white, and grey box XAI technology.

3.1 Conceptual model linking XAI characteristics to agricultural decision-making requirements

The primary goal of XAI is to obtain a human-interpretable model that answers questions like, "How does the model work?", "Why did it just provide this particular result?", and "What mistakes can it make?" (Hoffman et al. 2018). There are several terminologies, such as interpretability, transparency, fidelity, Scalability, etc. which trigger humans' curiosity about XAI. Besides, XAI methods can justify a decision, control the model, improve results, and discover new facts (Arrieta 2020).

For explainable AI (XAI) to work in agriculture, it needs to meet the special needs of the industry and its unique problems. Timeliness is very important since decisions like whether

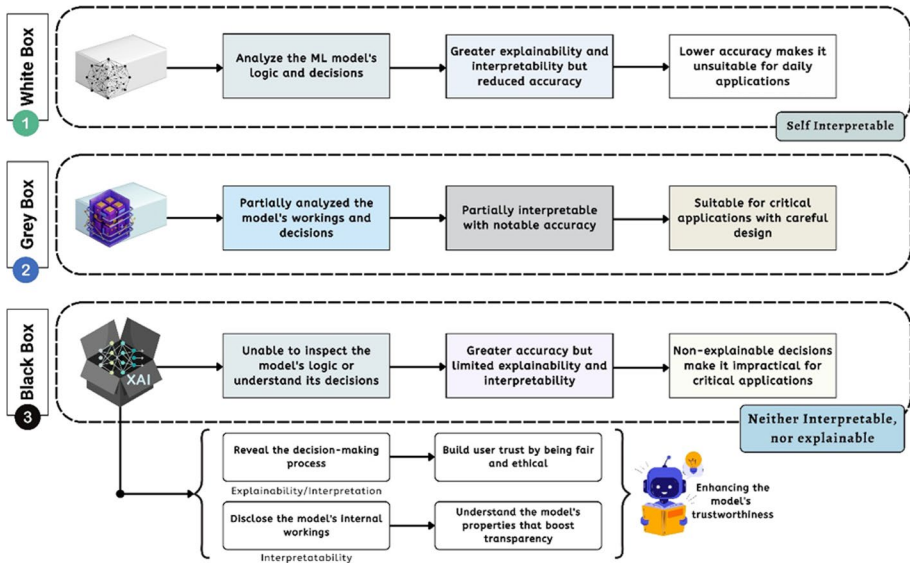


Fig. 4 A comparison of white-box, gray-box, and black-box models (Ali 2023)

to water or how to get rid of pests need quick answers to make the best use of resources and keep crops healthy. Usability is just as important, making sure that XAI explanations are clear and easy for non-technical users to understand, including farmers who don't know much about AI. This lets people act on insights without needing to know a lot about technology. For farmers to use XAI outcomes in important decisions that affect yields and sustainability, they need to trust them. Trust is a key part of adoption. Finally, context-specificity makes sure that XAI solutions take into consideration local factors like soil type, climate, and regional farming techniques. This makes explanations useful and relevant in a wide range of agricultural situations (Keskinbora 2019).

To improve the use of explainable AI (XAI) in sustainable agriculture, we need a conceptual model that connects the features of XAI to the needs of agricultural decision-making. interpretability directly enhances usability by offering straightforward, visual elucidations, such as feature importance plots, which facilitate farmers' understanding of crop production estimates and empower them to make informed decisions without necessitating technical proficiency. Transparency builds trust by providing detailed records of how models work, which gives farmers peace of mind that XAI outputs are reliable in high-stakes situations like pest control or planning irrigation. Also, scalability makes context-specificity better since XAI models that can handle different datasets may work in different agricultural settings, such different soil types, climates, or regional practices. This makes sure that the insights are useful and relevant for a wide range of farming situations. The overall mapping between XAI characteristics and decision-making requirements is presented in Fig. 5.

3.2 Taxonomy of XAI applications specific to agriculture

A taxonomy can be created by grouping explainable AI (XAI) applications into important agricultural areas and connecting them to certain activities and approaches which can help

		Decision-making Requirements			
XAI Characteristics		Interpretability	Transparency	Fidelity	Scalability
	Timeliness	Enables quick understanding of model outputs	Helps identify reasons for model decisions	Builds confidence in output accuracy	Ensures consistent performance as applications grow
	Usability	Enhances model interpretability for faster analysis	Clarifies model logic for effective use	Supports reliable outputs in varied contexts	Adapts to different scales and domains
	Trust	Fosters trust through explainable processes	Increases confidence in model outputs	Ensures reliability of decisions	Maintains trust across various settings
	Context-Specificity	Facilitates adaptation to specific scenarios	Ensures model decisions are contextually relevant	Provides dependable outputs for particular settings	Supports application across diverse contexts

Fig. 5 How XAI characteristics support decision making requirements

Table 1 Agricultural domain applications of XAI techniques

Agricultural domain	Specific task	XAI technique	Application example
Precision agriculture	Yield prediction	SHAP	Explaining key factors (e.g., soil pH)
Resource management	Irrigation optimization	LIME	Local explanations for water needs
Pest and disease management	Pest detection	Rule-based models	Rule sets for pest identification

organize the uses of XAI in sustainable agriculture in a systematic way. Precision agriculture (which includes monitoring crops, predicting yields, and analyzing soil), resource management (which focuses on optimizing water use and applying fertilizer), pest and disease management (which includes detection and mitigation strategies), supply chain optimization (which includes predicting market trends and planning logistics), and climate adaptation (which includes forecasting weather and planning for resilience) are all relevant areas. In each domain, certain tasks are matched with the right XAI approaches. For example, SHAP is used to find out which features are most important for predicting maize production, LIME is used to give local explanations for optimizing irrigation, rule-based models are used to find pests, and attention mechanisms are used to predict the weather (Ribeiro et al. 2016; Ai 2019; Cath et al. 2018). This taxonomy is organized in a hierarchy. This makes it easy to understand and improve XAI's role in agriculture. Table 1 provides the summary of the taxonomy of XAI in the context of agriculture.

Besides, it is necessary to understand the type of explanation and who benefits from XAI. In our study, we identified three types of explanations depicted in Fig. 6. In this figure, explanations can be classified into three types of models, such as:

- **Transparency-based explanations:** The notion of transparency corresponds to the concept of “full technical transparency” (Lords 2018). Both sources agree that models

beyond decision trees or rule bases cannot be fully transparent. Explaining the system this way may make it seem inconsistent, unreliable, “impenetrable,” or “foreign.”

- **Post-hoc explanations:** Popular after-the-fact explanations include approximation using a local model illustrating as a saliency map or a decision tree in LIME. Therefore, it is particularly beneficial for computer users and ethics students (Koh and Liang 2017). Methods include finding training set examples most relevant to a result and using case-based reasoning to find related instances (Montavon et al. 2017).
- **Layered explanations:** Following the previous discussion, multiple explanations for farmers may make sense. Can a composite explanation object include all the information needed to satisfy farmer and be unpacked (via accessor methods) to meet a recipient’s needs? The object has following layers:

Level 1 Identifies the agricultural domain: Precision agriculture

Level 2 Specifies the task: Yield prediction

Level 3 XAI Technique and Application: SHAP for feature importance in maize yield models

4 Challenges and opportunities of XAI in a multidisciplinary perspective

This section presents the challenges and opportunities of XAI from a multidisciplinary perspective. The section is classified into two subsections. Section 4.1 discusses the core cross-domain challenges for XAI, and Sect. 4.2 illustrates the Domain-Specific Opportunities of XAI.

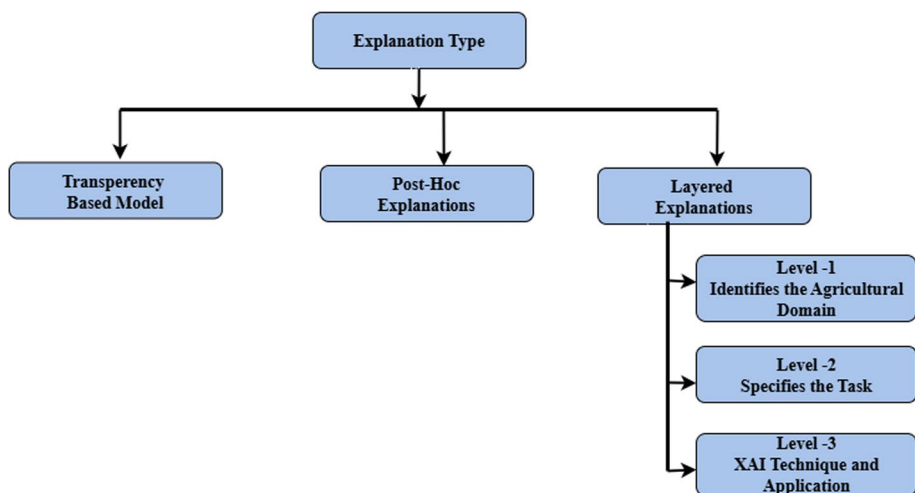


Fig. 6 A taxonomy of various explanatory models and methods in XAI (Das and Rad 2020)

4.1 Cross-domain challenges

Due to XAI's unsolved concerns, merging machine learning with explanations is tough. Balance between accuracy and interpretability is another important issue in explainable AI development. AI systems, especially deep neural networks, are changing how we do real-world activities. Due to their black-box nature, deep neural networks might create ethical and judicial problems and engender distrust in mission-critical applications. Due to data task diversity, XAI cross-domain difficulties develop. However, interpretability must be satisfied in healthcare and other applications. In medicine, a successful solution may not work in finance or autonomous systems. In many cases, a model trained for one domain fails in another. On the other hand, regularity and ethics in diverse fields might impact XAI success.

Learning about XAI has grown in popularity. Their interest has grown, and even governments are interested, mostly due to the European General Data Protection Regulation. The popularity of XAI stems from the recognition of ethics (Etzioni and Etzioni 2017; Bostrom and Yudkowsky 2018), trust, and prejudice (Stahl and Wright 2018; Weld and Bansal 2019; Lui and Lamb 2018; Hengstler et al. 2016; Chen 2019). People want fair, transparent, and issue-free AI systems (Challen et al. 2019; Sinz et al. 2019). Additionally, individuals are anxious about the result of adversarial instances in decision-making (Osoba and Welser 2017; Kurakin et al. 2016; Su et al. 2019; Huang et al. 2017). Some inputs are intended to fool AI systems into bad choices. Like tricking the AI into believing something's wrong. A research by Miller (Miller 2019) indicates that people seek explanations for AI judgments and want to understand how they are produced. Consistent explanations should provide comparable causes for similar facts (Sokol and Flach 2020). The AI system should explain its choice in the same circumstance. Explainable AI struggles with transparency. People desire more reliable and transparent AI. They want AI systems to explain their conclusions clearly and reliably. Alchemy AI works opaquely and produces unexpected results. People are not interested in AI-like alchemy (Tony 2024).

While AI has greatly improved the agriculture industry, its effect on agrarian operations is still limited compared to its potential and impacts in other sectors (Ahngar 2022). Deep learning models provide a substantial problem in data labeling (Alibabaei 2022). Deep learning is costly and requires significant computer power, making it challenging to utilize in rural locations with limited resources (Sourav and Emanuel 2021). However, recognizing diseases is difficult when they share symptoms and patterns (Barbedo 2016).

4.2 Domain-specific opportunities

XAI provides numerous good chances in specialized "domains." It may help clinicians comprehend how trained models make health and diagnostic judgments, making them more likely to trust and employ them. It also helps consumers feel treated properly in finance by explaining loan approvals and denials. Besides, XAI can explain why a self-driving vehicle stops or turns, making the system safer. XAI can explain how academic achievement is recorded, helping instructors assist students. These examples demonstrate how XAI enhances AI's usefulness, transparency, and trustworthiness in several aspects of life (Arrieta 2020). Several XAI approaches may be improved:

- If used properly, XAI helps us detect and avoid hostile cases. We can identify AI deci-

sion-making elements if the process is clear. Finding high-quality algorithms that can withstand assaults and give transparency is challenging (Kurakin et al. 2016). AI models should comprehend and explain their judgments via text or visual reports. In this method, we can enhance AI decision confidence.

- AI transparency improves user, stakeholder, and public confidence. Understanding how AI systems make judgments helps users trust, confidence and accept them (Rossi 2018). Understanding the rationale behind a choice is crucial for creating confidence in AI systems, regardless of the end-user's role (e.g., expert, developer, legislator, or ordinary person) (Arrieta 2020; King et al. 2020).
- Advanced and confident choices may be made using AI system explanations. Users may evaluate output dependability and validity. They are able to track and assess the link between input data and AI system output predictions using XAI approaches. Understanding the relationship between input data attributes and AI decision outcomes for various groups or categories is straightforward (Du et al. 2020).
- In agriculture, Explainable AI (XAI) presents opportunities for enhancing crop yield and improving the prediction of plant diseases (Muthu et al. 2023). Additionally, Transparency and explainability of models will build trust among farmers (Gardezi 2024).

Finally, these opportunities highlight the importance of transparency in explainable AI and build trust and concern about ethics, accountability, decision-making, and collaboration.

5 Multidomain applications of Explainable AI (XAI)

The multidomain applications show how XAI makes AI systems more intelligible, dependable, and useful in varied real-world contexts. It enables physicians and medical workers generate AI-driven diagnosis and suggestions, improving patient care. Second, in finance, XAI discusses credit scoring, fraud detection, and investment techniques for justice and accountability. XAI reveals why autonomous systems like automobiles and drones perform certain behaviors, boosting safety and user trust. It can detect and explain cybersecurity dangers, helping specialists react. XAI analyzes student performance and gives instructors and students meaningful feedback.

5.1 Medicine and healthcare: enhancing diagnostics and patient care

In medicine and healthcare, XAI provides explicit reasons for AI systems' prognoses and resolves, boosting their credibility and reliability. This approach is crucial in healthcare, as autonomous systems must provide responsible results (šimić 2021). AI-powered diagnostic tools have been utilized in modern labs for a long time (Amann et al. 2020). However, the incapacity of these systems to explain choices is a major issue. An AI-driven neural network created in the mid-1990s to determine hospital admission for pneumonia patients inadvertently concluded, based on its training, that asthmatics with pneumonia had a lower mortality risk, recommending against hospitalization. This contradicts the medical practice of admitting asthmatic pneumonia patients to critical care units for enhanced treatment, which improves survival. To fulfill regulatory norms and get medical accreditation, medical AI frameworks, particularly Clinical Decision Support Systems (CDSSs), must undergo rigor-

ous validation procedures (Higgins and Madai 2020). Recent research (Zhang et al. 2017; Holzinger et al. 2017; Tonekaboni et al. 2019; Holzinger et al. 2019; Hossain et al. 2020; Khodabandehloo et al. 2021; Katuwal et al. 2016; Che et al. 2016) emphasizes the importance of explainability in medical AI. Deep learning approaches in AI have transformed healthcare by automating patient triage, diagnosis, prognosis, therapy formulation, and care management (Jiang 2017). Machine learning algorithms like decision trees, Random Forests, and GAMs provide interpretable XAI modalities, making them vital in medicine and healthcare (Deng et al. 2018; Pawar et al. 2020).

5.2 Transportation: building trust in autonomous systems

Sensor arrays enable the monitoring and identification of things on highways, a key component of Intelligent Transportation Systems (ITS) (Garg and Kaur 2023). ITS has issues such as optimizing vehicle routing, anticipating demand, predicting traffic flow, optimizing processes, estimating arrival times, and detecting abnormalities or security breaches (Tsolaki et al. 2023). Cooperative Intelligent Transportation Systems prioritize trust. The European Telecommunications Standards Institute (ETSI) emphasizes the need of trust in C-ITS security (ETSI 2021). On March 18, 2018, an automated automobile killed a pedestrian in Arizona, highlighting the challenge of establishing the role of artificial intelligence (AI) in such catastrophes (Wakabayashi 2018). The complexities of AI's functioning and accident causes remain unclear (Van Brummelen et al. 2018). Identifying the reasons of unpredictable or unexpected behaviors in Intelligent Connected Vehicles (ICVs) is challenging due to the opaqueness of "black box" AI solutions (Jaigirdar et al. 2020). In securing ICV networks, XAI is a revolutionary solution. For security threat detection and mitigation, an Intrusion Detection System (IDS) must be adaptable, automated, and explainable. XAI's clear and interpretable logic enables security experts to identify security flaws and intrusion attempts more accurately (Sarhan et al. 2021; Zebin et al. 2022; Marino et al. 2018; Das et al. 2021). In September 2022, researchers used XAI and CAN bus data to create a comprehensive vehicle security system. The "VisExp" technique uses XAI to visualize explanatory data, improving user understanding and confidence in the system's operational logic (Lundberg 2022).

5.3 Defense: ensuring accountability and strategic decision-making

AI algorithms play a significant role in defense, including speech recognition systems (Lotfidereshgi et al. 2018) and object detection and recognition to identify and classify items in operational environments (Pietrow and Matuszewski 2017). XAI was developed by military researchers and is widely studied in academia because to its usefulness in understanding AI-driven decision-making. Explaining AI methodology, especially in military operations, is difficult and frequently obscures AI's strategic decisions. Knight (MIT Technology Review) notes that deploying autonomous systems in combat contexts often leads to ethical and legal issues, particularly when making life-or-death judgments (Knight 2017). The complexity of AI applications in healthcare and other fields emphasizes the need for openness and accountability.

5.4 Education: personalizing learning with transparency

In education, AI has greatly improved human lives, triggering the ‘4th Industrial Revolution’ (World Economic Forum 2016; Timms 2016). A research on AI in Education Market Research highlights the worldwide market’s growth from \$1.1 billion in 2019 to \$25.7 billion by 2030, with a forecast worth of \$126 billion by 2025 (Market 2020; Liu 2020). This exponential expansion is reflected in the educational sector’s enhanced offers to the learning community, driven by EDM advances. Figure 7 illustrates a data-driven instructional model in which educators use data analytics to get deeper student performance insights. Chitti et al. (2020) demonstrate the importance of AI-based frameworks, particularly XAI components, in predicting students’ academic success. Alternatively, Avella et al. (2016) propose a “data-driven approach” using social media data visualization, predictive analytics, clustering, association mining, model creation, and data segmentation to improve educational decisions. This analytical rigor helps educators use ML models to predict student success and attrition rates, demonstrating the importance of AI/ML model explainability in boosting confidence and accelerating technological integration in education. Logistic Regression, Decision Trees, Support Vector Machines, Artificial Neural Networks, and Random Forests are common EDM techniques. However, these methods have been criticized for their opacity in previous studies (Ribeiro et al. 2016; Binh et al. 2017; Amrieh et al. 2016). Hasib et al. (2022a) created a secondary school student performance prediction system employing

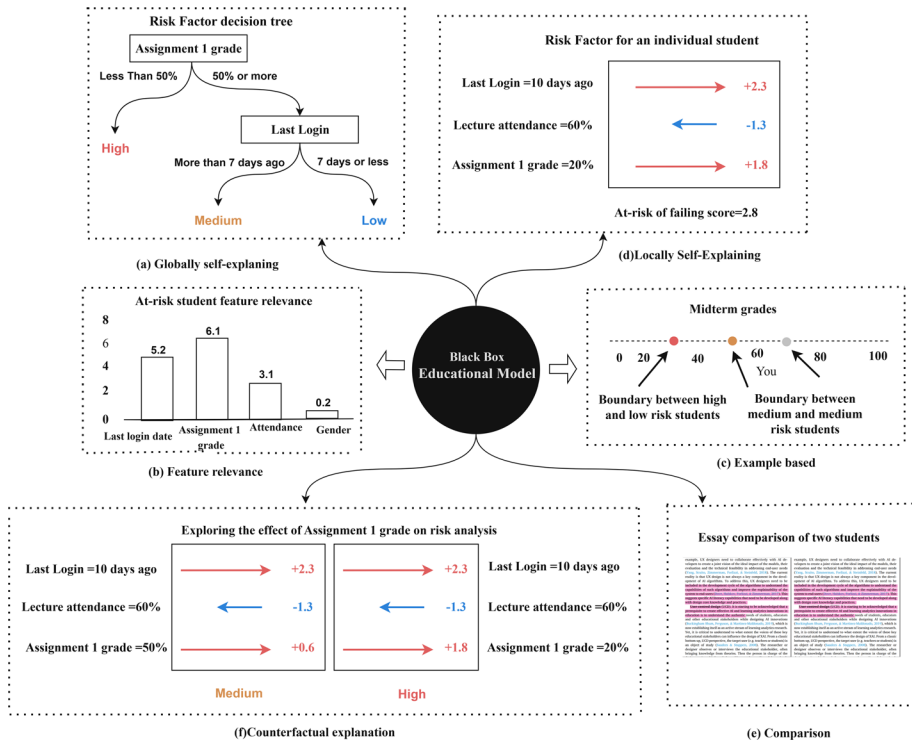


Fig. 7 Catalog of explainability methods utilized in AI for education: a sector-specific analysis (Khosravi 2022)

five ML classifiers and XAI for attribute correlation. Using a dataset from the University of Minho in Portugal, Al-Shehri (2017) shows that Support Vector Machine and K-nearest neighbor algorithms effectively predict student grades, with correlation values of 0.96 and 0.95, respectively. More study on high school student performance modeling using SVM, RF, KNN, and MLP classifiers for multi-label classification showed that SVM defined student data label spaces better. Text annotations by student mentors provide a nuanced view of student states (Jeon et al. 2019), emphasizing performance factors like GPA, evaluations, and demographics (Albreiki et al. 2021). Khosravi (2022) outline methods to improve explainability in AI, enabling stakeholders to gain more understanding of machine learning models. AIED applications include adaptive learning environments, tailored tests, automated feedback mechanisms, tutoring utilities, and learning analytics dashboards. This emphasizes the need for educators to have dashboards for model result interpretation, suggesting XAI approaches as a strategic option to dashboard construction.

5.5 Agriculture: optimizing farming with interpretable AI

In 2018, agriculture contributed to approximately 4% of the global GDP, with nearly 28% of the worldwide populace tasked with feeding 1 billion individuals engaged in agricultural endeavours (World Bank Group 2024; Hasib et al. 2022b). The escalation in food demand, primarily attributed to population growth, necessitates enhanced farm productivity and crop protection as imperative measures (Orchi et al. 2022). The prevalence of crop diseases, adversely affecting the yield and quality of agricultural produce, represents a significant challenge within the farming sector (Ouhami et al. 2021).

Mastitis, a multibillion-dollar dairy disease, raises antibiotic resistance worries. Ryan et al. (2021) used a counterfactual methodology, diverging from XAI, to develop a machine learning framework to predict bovine mastitis and provide interpretative explanations. By involving stakeholders, the research revealed which components of the system's forecasts might be pragmatically applied, demonstrating the model's ability to give considerable explanations with minimal data set changes. Delaney et al. (2022) showed how AI can anticipate long-term milk production, improve time series analysis with data-augmentation procedures, and use KNN approaches from Case-Based Reasoning (CBR). CBR's prefactual explanations help farmers plan future milk yield increases, bridging the gap between predictive analytics and actionable information in the dairy business. Early crop disease diagnosis and control are crucial for farmers, stakeholders, and precision agriculture researchers to reduce production deficits. Traditional agrarian methods rely on farm workers manually diagnosing plant diseases and professional advice using microscopy. Paudel et al. (2023) demonstrate how neural networks independently identify essential features, enabling accurate crop yield predictions. Automating feature extraction reduces manual feature specification, which is essential for crop yield predictions.

A new hybrid LSTM-XGB model uses SHAP for yield estimates via feature contribution analysis. Chi et al. (2023) used Raman spectroscopy and Machine Learning to classify cottonseed cultivars accurately. This demonstrates the XGBoost model's high accuracy in cottonseed identification and the SHAP framework's clarity in seed classification. Osibo (2023) introduced a hybrid LSTM-XGB model for wheat yield prediction that outperformed SOTA models. SHAP tool analysis clarified LSTM feature extraction and XGB training, highlighting feature contribution to yield predictions and improving model interpretability and appli-

cation in agriculture. Deep Learning (DL) has revolutionized crop classification (Kamrul et al. 2020), fruit grading (Hossain et al. 2018), pest detection (Türkolu 2019), plant disease identification (Trivedi 2021), and weed discernment (Mishra 2022). Wei (2022) investigated the internal interpretability of DL models for fruit leaf disease prediction using an advanced ResNet-Attention model with the attention mechanism. Using three interpretative algorithms—SmoothGrad (Smilkov et al. 2017), LIME (Ribeiro et al. 2016), and Grad-CAM (Selvaraju et al. 2017)—they found that model feature extraction performance differs across agricultural categorization tasks. Nahiduzzaman (2023) also introduced an Explainable Artificial Intelligence (XAI) framework that uses the PDS-CNN model and SHAP for XAI to analyze specific image sections for mulberry leaf disease classification, improving categorization. Bhandari et al. (2023) introduced BotanicX-AI, an XAI-driven deep-learning system for tomato leaf disease recognition, using EfficientNetB5 and transfer learning. GradCAM and LIME provided significant model explanation and interpretability insights. Mehedi (2022) used a transfer learning strategy with the EfficientNetV2L model and XAI, specifically LIME, to make sure that finding plant diseases was reliable and transparent. Khalid and Karan (2024) showed that transfer learning improves model efficacy by using pre-existing information and integrating XAI via GradCAM to visualize disease signals in plant imagery. Gilbert et al. (2023) developed XAI models for Black Sigatoka detection in banana plants using LIME and Integrated Gradients to create transparent and interpretable models, demonstrating the dynamic junction of sophisticated machine learning with agricultural innovation.

6 Results synthesis and discussion

Agriculture is vital for a country's development, economy, and well-being. According to the UN Food and Agriculture Organization, the world population will exceed 9 billion by 2050 (Liu 2020). To cope with this large population, world agricultural productivity must be increased. In this regard, the advancement of AI, ML, and DL in agriculture is expanding daily. These technologies provide numerous benefits in the field of agriculture. For instance, they increase crop productivity, monitor crop health, disease recognition, crop yield identification, cattle identification, fruit counting, weather forecasting, and anticipate crop harvest readiness. Altogether, these technologies make farming more efficient by automating key tasks, helping farmers make better decisions about irrigation and fertilization, and improving overall farm management. In this section, we reviewed a few papers that used ML and DL in the agriculture field and examined the challenges and limitations that we face while using ML and DL in the agriculture field. Figure 8 shows the application of DL approaches with XAI in the agriculture sector in the fruit grading system.

6.1 Traditional machine learning vs deep learning: insights from existing studies

The incorporation of ML models in the agriculture field is increasing daily. Researchers are striving to develop automated systems to enhance crop production. El Bilali et al. (2021) evaluated the performance of machine learning models, including Random Forest (RF), Support Vector Regression (SVR), Artificial Neural Networks (ANN), and AdaBoost, for predicting irrigation water quality parameters. The main purpose of this study was to evalu-

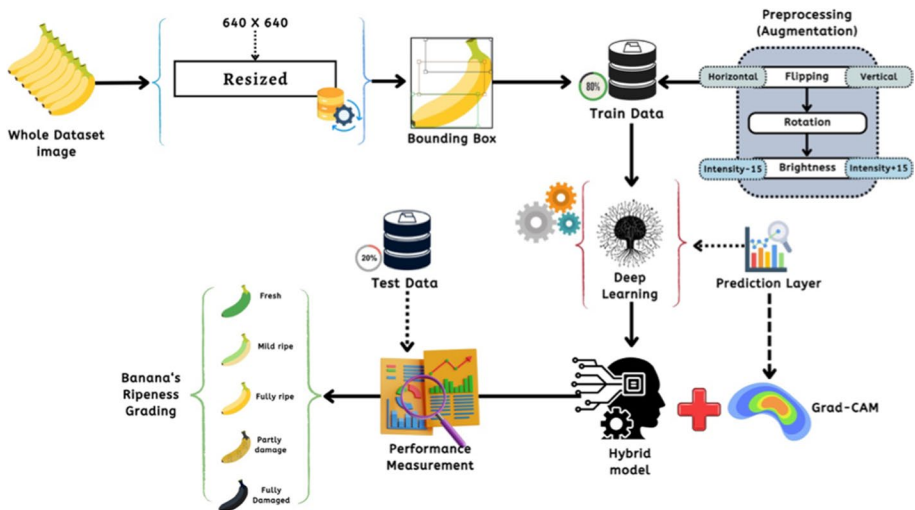


Fig. 8 A deep learning scheme for fruit grading system with XAI (Hossain et al. 2018)

ate whether the ML model can effectively forecast the groundwater quality for irrigation purposes. The study utilized 520 data samples collected from the River Basin Agency of Bouregreg and Chaouia. The ANN, RF, and AdaBoost models demonstrated high accuracy in predicting TDS parameters, achieving a correlation coefficient (r) of 0.95, a Root Mean Square Error (RMSE) of 457 mg/L, and an average Relative Bias in Absolute Value (RBIAS) of 1%. However, the study's limitations include using a restricted dataset and a narrowly defined temporal scope. Similarly, Shrivastava and Pradhan (2021) employed seven machine learning approaches for classifying rice plant diseases. The research focused on using color feature extraction methods for rapid diagnosis and intervention to prevent agricultural economic losses. The dataset consisted of 619 images categorized into four classes. Among the classifiers, SVM achieved the highest accuracy of 94.65%. Nevertheless, the study's shortcomings include its exclusive reliance on color features, which may overlook other relevant information, and the limited dataset, which reduces the model's generalizability to different crops and scenarios.

Alagumariappan et al. (2020) developed a decision support system using Raspberry PI to classify plant health. They used an Extreme Learning Machine (ELM) and SVM with linear Kernel (LSVM) and SVM with polynomial kernels (PSVM) to analyze a dataset of 200 images from three leaf types. The system achieved an accuracy of 95% using ELM, demonstrating its effectiveness in real-time plant disease diagnosis. Iqbal and Talukder (2020) have developed an automated framework to detect potato leaf disease. They processed 450 photos using the HSV color space to distinguish between healthy and affected areas of the leaves. This segmentation process enables the accurate extraction of features and classification. It utilizes seven classifiers and achieves a high accuracy rate of 97% using the RF algorithm. The dataset used includes 14 different crops from the Plant Village database. The combined findings of this research emphasize the significant advancements in agricultural disease management technology and the possibilities for using these advanced systems to detect and prevent diseases before they occur. Shakil (2023) developed an automated dragon fruit illness detection system using ANOVA and LASSO to identify feature significance. Six

machine learning classifiers were carefully applied to the top feature sets and evaluated across seven criteria. AdaBoost and Random Forest (RF) classifiers under the LASSO feature selection paradigm achieved an unmatched accuracy of 96.29%. This study highlights the revolutionary significance of machine learning in agricultural disease identification, providing crucial insights for better diagnostic treatments. A web-based machine vision expert system developed by Rajbongshi et al. (2022) allows farmers to diagnose cauliflower problems using smartphone or handheld imagery. Using seven performance metrics, the study used k-means clustering to segment disease-afflicted zones and statistical and co-occurrence feature extraction on 776 images to evaluate six algorithms (BayesNet, Kstar, RF, LMT, BPN, and J48). RF was the most accurate at 89.00%. In addition, Rajbongshi et al. (2021) created an autonomous entity that uses k-means clustering to identify sunflower disorders using computer vision. The work extracts GLCM and statistical features from 650 photos of various sunflower diseases. Five classifiers (Logistic Regression, Naïve Bayes, RF, J48, and K-star) were used for disease classification. The RF classifier had the most incredible mean accuracy of 90.68%, opening up new agricultural protocols for proactive identification and management.

Besides, using a machine learning model to classify leaves and fruit is essential to agriculture. An innovative machine vision-based system was developed by Qadri (2019), and their team was a game-changer for the agricultural industry. With an accuracy of 98.14%, the system can classify citrus leaves into eight different varieties using a dataset of 1,600 images. By utilizing techniques such as threshold-oriented clustering, segmentation, and feature fusion, including texture, binary, and spectral features, the system provides a robust method for classification based on leaf characteristics. A total of 57 features were initially acquired, with 15 optimized features selected using the supervised correlation-based feature selection (CFS) technique. They demonstrated the immense potential of using machine vision in agriculture, paving the way for future breakthroughs in the field. On the contrary, Biswas et al. (2021) employed RF, AdaBoost, DT, Gradient Boosting, and Bagging classifiers to detect carrot diseases. K-means clustering was utilized to segment the disease-affected areas. Among the five classifiers, RF outperformed the others, achieving an accuracy of 94.17% using nine features for model training and testing. However, the study's limitation lies in using a restricted number of features, which may impact the consistency and overall quality of the results. Similarly, using field-level data, Habib et al. (2020) developed a system for recognizing papaya diseases. They applied k-means clustering to segment the affected regions, extracted two types of features, and used three machine learning classifiers to classify the disease. This study's limitation is that their dataset was very limited in small size and the limited number of machine learning classifiers were employed, which may affect the robustness and generalizability of the results.

Conversely, Koklu et al. (2021) used machine learning models to classify two pumpkin seeds based on their physical characteristics. They tested five models using 1,300 Çerçevelik and 1,200 U?urp Sivrisi seeds. The Support Vector Machine model was the most accurate, with an 88.64% success rate. Beyond applying five distinct machine learning models, the research meticulously evaluated model performance using a 10-fold cross-validation method, ensuring the robustness of the findings. This comprehensive methodology highlights the potential of machine learning in agricultural classification and sets a precedent for future research in precision agriculture and seed quality assessment.

Moreover, crop prediction using a machine learning model can improve crop yield and reduce the use of materials such as fertilizers, insecticides, and water in agriculture. Hence, Wei et al. (2020) developed a model using a random forest regression classifier that uses satellite imagery and on-farm sampling data to predict carrot yield. The model uses high-resolution spectral bands that change over time to achieve an error of 2.7 Mg ha⁻¹ and an R² of 0.82. This shows that the model is effective in forecasting yields in commercial fields. Nonetheless, it is different from using vegetation indices and demonstrates the potential of machine learning to identify new patterns in agricultural data. As a result, this approach supports precision agriculture objectives by enabling accurate yield mapping and management strategies. Additionally, Kuradusenge (2023) have conducted a study on crop yield prediction for Irish potatoes and maize in Rwanda, utilizing machine learning models. They have analyzed historical weather and yield data and compared the performance of RF, Polynomial Regression (PR), and Support Vector Regression (SVR) models. Their analysis, based on data from 2006 to 2021, has demonstrated that the RF model outperforms the other models. Experimental result shows that the RF achieved a RMSE of 510.8 for potato yields and 129.9 for maize yields, while a R² of 0.875 for potato yields and 0.817 for maize yields. This finding could have significant implications for decision-making and agricultural practices, particularly regarding climate variability. Then, Elbasi (2023) conducted a detailed analysis of 15 machine learning algorithms for agriculture, focusing on crop recommendation. The study was based on a dataset of 2200 records across 22 crop types, and it placed significant emphasis on data collection from IoT devices. The research team applied various models such as Bayes Net (BN), Naïve Bayes (NB), Logistic Regression (LR), Multilayer-Perception (MLP), Simple Logistic (SL), IBK, KSTAR, LWL, AdaBoostM1, Regression, Decision Tree (DT), Hoeffding Tree (HT), J48, Random Forest (RF), Random Tree (RT), etc to optimize crop selection, and the results were impressive. Notably, the Bayes Net algorithm achieved a classification accuracy of 99.59%, highlighting the immense potential of machine learning in enhancing agricultural productivity and decision-making. On the other hand, the research conducted by Rajpal (2020) detected Black Rot fungal disease in grapevines using color-based segmentation techniques. The classification of segmented images was performed using various kernels of the SVM, including linear, RBF, and polynomial. The dataset used consisted of 400 images. Among the kernels, the RBF kernel achieved the highest diagnostic accuracy of 94.1%. However, the limited size of the dataset may hinder the generalization of results to diverse environmental conditions. Additionally, variations in lighting conditions and leaf color could affect the effectiveness of the HSV and L*a*b color models. Similarly, Chakraborty et al. (2021) employed a multiclass SVM to detect infections in apple leaves. Their dataset included 500 images representing eight apple species, with classes for leaves affected by Black Rot, Cedar Apple Rust, and healthy leaves. The study achieved impressive results, with an average Precision, Recall, and F1-score of 96% across all classes. Nonetheless, certain limitations were noted. The exclusive use of the SVM model may not be the optimal choice compared to more advanced machine learning models, and the relatively small dataset could limit the model's robustness and generalizability to broader scenarios. Besides, M. Cockburn's (2020) review of ML in dairy farm management covers studies published between 2015 and 2020, focusing on integrating large datasets from various sources such as herd management systems, behavioral sensors, feeding lists, breeding schedules, and health records.

Based on the above discussion, the review highlights the potential and challenges of using ML algorithms to improve dairy farming practices, as presented in Table 2. However, the review emphasizes the need for more comprehensive and integrated datasets to enhance ML algorithms' reliability and practical application in farming scenarios. On the other hand, Fig. 9 shows the statistical data visualization of the papers who worked on the agriculture with ML and XAI in terms of their maximum accuracy and used sample dataset size.

Besides, crop yield is an important field where DL is extensively used. The DL models need a lot of images to make a system work effectively. Guillén (2021) utilized deep learning (DL) techniques for frost prediction in agriculture, a critical measure for preventing crop losses. While Jeong et al. (2022) proposed a remote-sensing integrated crop model (RSCM) to generate precise pixel-scale reference yields. The authors tested five different deep-learning structures and found that the hybrid structure of LSTM and 1D-CNN layers performs the best for predicting rice yield about two months ahead of harvest time. Alibabaei et al. (2021) compared their proposed methods with traditional machine learning methods like Convolutional Neural Networks, Multi-Layer Perceptrons, and Random Forest Regression, where the Bidirectional LSTM achieved the best performance compared to other models. This study also highlights the potential of deep learning models in improving accuracy and decision-making in agricultural yield prediction. While de Camargo et al. (2021) proposed an optimized model based on ResNet-18 for classifying weed and crop plants using UAV images. The model's performance was evaluated using a classification map generated for each UAV test image. The accuracy of the models varies based on the filter configuration used. The overall classification accuracy of the models was found to be 94%, indicating strong predictive capabilities. Ullah et al. (2022) focused on the application based on deep learning models in the automation of pest identification, addressing the need for more accurate systems. The DeepPestNet model achieves an optimal accuracy of 100% on Deng's crops dataset for pest recognition.

On the other hand, Fruit disease detection using DL is essential as it helps farmers identify and manage diseases early, preventing crop losses. A system is developed for estimating various fruit types, maturity, and weight using image processing technologies (Faisal et al. 2020). Four pre-trained CNN architectures are used for date type and maturity estimation, and an SVM for weight estimation. The paper attained high accuracy rates for date type and maturity estimation, which is above 98%, and moderate accuracy for weight estimation, which is 84.3%. Kukreja et al. (2020) employed various image processing methods to enhance image quality and maintain consistent data for training and validation. They used a dense CNN to detect and classify based on image data. The applied model is compared with a dense model where data augmentation and pre-processing techniques were not used. Besides, a combined approach using a fine-tuned, pre-trained deep learning model (VGG19) for fruit disease classification is proposed by Nasir (2021). Aims to improve image-based fruit classification by utilizing CNN models. The proposed method achieved an accuracy of up to 99.6% on fruit disease classification compared to other methods that were used previously. A technique that optimally utilized deep fusion methodology was introduced, and selected the most essential features to achieve an improved classification outcome score that is 97.8% (Khan et al. 2020). The aim is to use computer vision (CV) techniques to detect early-stage diseases.

Again, DL algorithms can also accurately identify and classify various types of leaf diseases, providing precise insights that help farmers in the agricultural field. An accurate

Table 2 Summary on ML in agricultural disease detection

References	Domain	Dataset	ML algorithm	Accuracy	Segmentation	Limitations
El Bilali et al. (2021)	Water quality	520 images	RF, SVR, ANN, AdaBoost	Adaboost RMSE—457 mg/L	No	Limited dataset size and time range Not generalizable to all areas
Shrivastava and Pradhan (2021)	Rice plant diseases	619 images	SVM, DC, KNN, NB, DT, RF, LR	SVM—94.65%, DC—92.34%, KNN—91.39%, NB—75.72%, DT—83.18%, RF—92.52%, LR—75.85%	No	Limited dataset size Relies on color features for classification, which may not capture all relevant information about the disease
Alagumariappan et al. (2020)	Plant diseases	200 images	LSVM, PSVM, ELM	LSVM—87.5%, PSVM—92.5%, ELM—95%	Yes	Focused on a limited number of plant types and diseases Requires further validation on a broader range of diseases and conditions
Iqbal and Talukder (2020)	Potato leaf diseases	450 images	RF, LR, KNN, DT, NB, LDA, and SVM	RF—97%, LR—94%, KNN—91%, DT—91%, NB—84%, LDA—78%, and SVM—37%	Yes	Specific to Early Blight and Late Blight diseases Limited by the dataset's diversity and image quality
Shakil (2023)	Dragon fruit	404 images	AdaBoost, RF, DT, KNN, LR, SVM	AdaBoost, RF—96.29%	Yes	Applicable only to dragon fruit diseases Relies on real-time picture data quality and variance LASSO feature set-focused

Table 2 (continued)

References	Domain	Dataset	ML algorithm	Accuracy	Segmentation	Limitations
Rajbongshi et al. (2022)	cauliflower disease	776 images	BayesNet, Kstar, RF, LMT, BPN, and J48	Bayes-Net—80.06%, Kstar—88.27%, RF—88.61%, LMT—87.67%, BPN—85.66%, J48—86.44%	Yes	Concentrated on four specific forms of cauliflower diseases Accurate illness detection relies on the quality of the photograph and the environmental factors Additional testing and validation are necessary across various illness types and settings
Rajbongshi et al. (2021)	Sunflowers	650 images	LR, NB, RF, J48, K-star	LR—88.85%, NB—75.21%, RF—90.68%, J48—78.90%, K-star—81.33%	Yes	Focused on four sunflower diseases Dependent on image quality Needs broader validation
Qadri (2019)	Citrus varieties	1600 images	MLP, RF, J48, NB	MLP—98.14%, RF—81.75%, J4—880.34%, NB—75.70%	Yes	Reliance on high-quality, controlled imaging conditions Classification performance may vary with changes in environmental conditions or leaf quality
Biswas et al. (2021)	Carrot diseases	599 images	RF, Ada-Boost, DT, GB, and Bagging	RF—94.17%, Ada-boost—84.17%, DT—88.33%, GB—87.50%, Bagging 92.50%	Yes	Limited for carrot disease only Only 10 features were extracted, while the impact of other features was not explored
Habib et al. (2020)	Papaya diseases	126 images	SVM, DT, NB	SVM—92.15%	Yes	Limited to papaya fruits, not generalizable to other types of fruits Confined image data

Table 2 (continued)

References	Domain	Dataset	ML algorithm	Accuracy	Segmentation	Limitations
Koklu et al. (2021)	Pumpkin seed	2500 images	LR, MLP, SVM, RF, DT, and kNN	LR—87.92%, MLP—88.52%, SVM—88.64%, RF—87.56%, DT—87.64%, kNN—87.64%	Yes	Focuses on only two varieties which limiting the study's applicability to a broader range of pumpkin seeds Relies on high-quality image acquisition and preprocessing, which might not be feasible in all practical scenarios
Wei et al. (2020)	Carrot yield	15,093 samples	RF	R^2 value of 0.82, RMSE of 2.64, MAE of 1.74	Yes	Limited to carrot crops, might not directly apply to other horticultural or crops Performance depends on the quality of satellite images and accurate yield sampling
Kura-dusenge (2023)	Irish potatoes	Data from 2006 to 2021	RF, PR, and SVR	Potato—RMSE—510.8 and maize—RMSE—129.9	No	Limited to rainfall and temperature data Specific to Masanaga district Other weather variables like humidity, soil moisture, are not included

Table 2 (continued)

References	Domain	Dataset	ML algorithm	Accuracy	Segmentation	Limitations
Elbasi (2023)	Crop production	2200 data	BN, NB, LR, MLP, SL, IBK, KSTAR, LWL, Ada-BoostM1, Regression, DT, HT, J48, RF, RT	BN—99.59%, NB—99.46%, LR—97.99%, MLP—98.79%, SL—98.66%, IBK—97.86%, KSTAR—97.86%, LWL—76.74%, Ada-BoostM1—6.82%, Regression—98.38%, DT—88.50%, HT—99.46%, J48—98.79%, RF—99.46%, RT—98.12%	No	Limited to certain features like nitrogen content, temperature, pH, rainfall, etc. May not cover all crop types or geographical variations
Rajpal (2020)	Black Rot disease in Grape Vines	400 images	SVM	Linear Kernel—93.3%, RBF Kernel—94.1%, Polynomial Kernel—93.9%	Yes	Higher dependency on color models impacted the efficiency of results The use of only one classifier restricts the acceptance and generalization of the findings
Chakraborty et al. (2021)	Black Rot and Cedar Apple Rust diseases in plants	500 images	SVM	SVM-96% (Precision, Recall, F1-score)	Yes	Reliance on only the SVM model might not always be optimal Limited size of dataset utilization
Cockburn (2020)	Dairy farms management	319,934 sample	K-NN, multilevel logistic regression, RF, boosted tree models, SVM, PCA	Not mentioned (NM)	No	Integration of diverse data sources remains challenging Most algorithms are not yet reliable for practical implementation

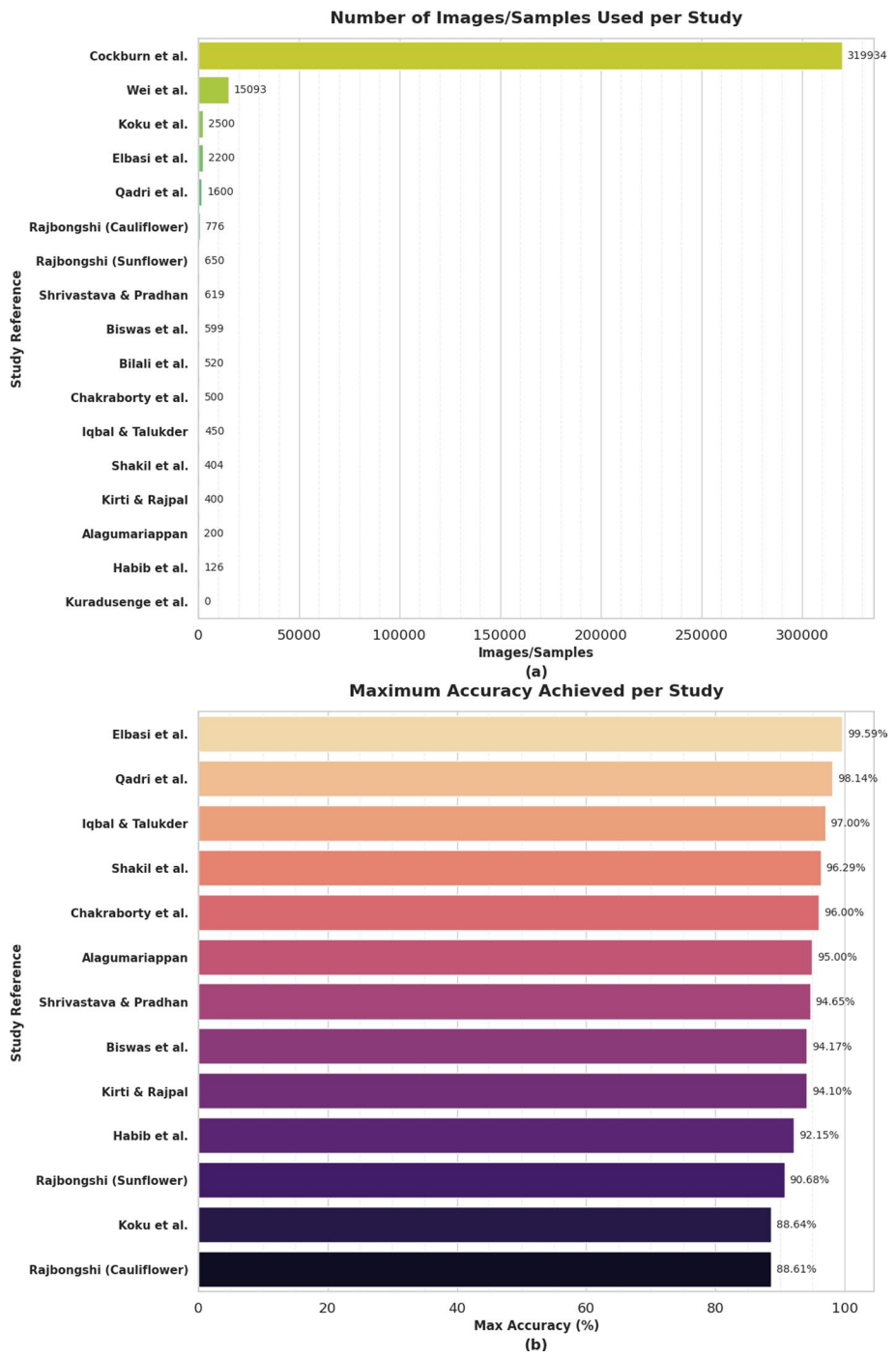


Fig. 9 A comparison among the papers that worked on ML and XAI in agriculture in terms of **a** used sample or dataset size and **b** maximum accuracy

identification approach for apple leaf diseases was suggested by Liu et al. (2017) based on AlexNet. The proposed CNN-based model achieved an accuracy of 97.62%, which is higher than other models, demonstrating its effectiveness in identifying apple leaf diseases. Also, Deep learning may be used to quickly recognize spot diseases in plants, stop diseases from spreading, and save crops. Fuentes et al. (2017) proposed a deep-learning approach for identifying various diseases and pests in tomato plants using images taken on-site with camera devices. It evaluated three main families of detectors, Faster R-CNN, R-FCN, and SSD, which are considered and combined with deep feature extractors like VGG net and ResNet. Hassan et al. (2021) implemented four different CNN models: InceptionV3, InceptionResNetV2, MobileNetV2, and EfficientNetB0, and compared their performance with others using machine-learning techniques. EfficientNetB0 achieved the best accuracy rate of 99.56%, which is a lightweight and efficient CNN model that considers network depth, width, and resolution. Ma et al. (2020) provided a methodology for enhancing the interpretability of viable and non-viable seeds in a deep learning approach. It utilized principal component analysis (PCA) and support vector machine (SVM) to extract the vital light absorption differences from the NIR-HSI data with high accuracy (90%). Haq (2022) proposed a novel CNN LVQ model that combines CNN and learning vector quantization (LVQ) to classify weed segments from UAV images. Three pre-trained CNN models, ResNet-50, Xception, and VGG16, were used for weed identification in potatoes, sunflowers, and maize crops. Among these, CNN LVQ models achieve an overall accuracy of 99.44% for weed detection, which is significantly higher than previous studies using ML and DL methods. A system that involves adapting a multirotor drone that analyzes the images captured by the drone's camera using CNNs is proposed by Rivas et al. (2018). However, the above studies have been presented in Table 3. The system was tested in a case study conducted on a cattle farm in Salamanca, Spain, and it achieved a high accuracy of 95.5% in identifying cattle from the images. On the other hand, Fig. 10 compares the papers that worked on the DL models with XAI in agriculture regarding the frequently used models or the most repeated number of models, and the highest accuracy they achieved while working with DL models.

7 Critical analysis and gaps

This literature section is for XAI in agriculture where many great contributors have made significant contributions. The following subsection depicts the relevant discussion with future directions.

7.1 Emerging research trends of XAI in agriculture

AI techniques, such as ML and DL, have become indispensable in the agricultural industry, significantly impacting crop yield prediction, plant and leaf disease detection, dairy farming optimization, and more. We reviewed several research studies that applied DL and ML in agriculture in the previous section. Our analysis revealed that while these approaches consistently achieved high levels of accuracy in their respective applications, their lack of transparency highlighted the need for improved explainability. In critical agricultural applications, explanatory power and interpretability are as important as accuracy. Therefore, there is a pressing need to explore and evaluate the role of Explainable Artificial Intelli-

Table 3 Summary on DL in agricultural disease detection

References	Domain	Dataset	Augmentation	DL Algorithm	Accuracy	Limitations
Guillén (2021)	Crops	Time series data generated by IoT infrastructure	NM	LSTM	RMSE = 0.6902	Utilized a 3-month dataset which does not represent long-term climate trend
Jeong et al. (2022)	Crops	Time-series data	NM	LSTM and 1D-CNN	$R^2 = 0.86$, RMSE = 0.61	Relies on satellite images which may not capture all factors affecting crop yields Not generalizable for all regions
Alibabaei et al. (2021)	Crops	Climate Dataset	NM	LSTM and Gated Recurrent Unit (GRU)	$R^2 = 0.97$ to 0.99	Only focuses on tomato and potato yield prediction
de Camargo et al. (2021)	Crops	Acquired during a UAV flight campaign	Yes	ResNet-18, DCNN classifier	ACC = 94%	Showed some confusion between similar weed species unknown species
Ullah et al. (2022)	Crops	1686 images	Yes	PestNet	ACC = 98.92%	Only compares DeepPestNet with two pre-trained deep learning models
Faisal et al. (2020)	Fruits	Used a dataset built by the Center of Smart Robotics Research	Yes	ResNet, VGG-19, Inception-V3, NASNet	ACC = 99.175%	Another DL method will be needed for segmentation the system needs extension to more fruit types
Kukreja et al. (2020)	Fruits	150 images increased to 1200 through augmentation	Yes	Dense CNN	ACC = 89.1%	only used a small dataset of 150 original images without augmentation, the model might not have performed effectively
Nasir (2021)	Fruits	Fruit 360 dataset (65,429 images)	Yes	VGG19	ACC = 99.6%	Multi-fruit classification features are missing

Table 3 (continued)

References	Domain	Dataset	Augmentation	DL Algorithm	Accuracy	Limitations
Khan et al. (2020)	Fruits	Plant Village dataset	Yes	DCNN models (AlexNet, VGG)	ACC = 97.8%	Only considers five types of diseases for three fruits
Liu et al. (2017)	Leaf	1053 images	Yes	AlexNet	ACC = 97.62%	More types of apple leaf diseases and high-quality natural images of apple leaf diseases still need to be gathered
Fuentes et al. (2017)	Plant	Tomato disease dataset (self-collected)	Yes	Faster R-CNN, SSD, single-shot multibox detector	ACC = NM	The system may confuse similar diseases dataset is limited in size
Hassan et al. (2021)	Plant	54,305 images	Yes	InceptionV3, Inception-ResNetV2, MobilenetV2, EfficientNetB0	ACC = 99.56%	Models may not generalize on new or real-time images
Ma et al. (2020)	Seed	Collected by authors	Yes	CNN	ACC—near 90%	Only evaluated on spinach seeds not tested on other valuable crops
Haq (2022)	Weed	“Weed detection in soybean crops” dataset	Yes	CNN	ACC = 99.44%	Needs to collect more images from different areas
Rivas et al. (2018)	Cattle	13,520 images	Yes	CNNs	ACC = 97.1%	Encounters challenges when same animal crosses the path multiple times

gence (XAI) in agriculture to meet the growing demands for trust, usability, and informed decision-making in this field. Integration of XAI can bridge the gap between complex AI models and end-users. Ultimately, this can lead to more sustainable, transparent, and data-informed agricultural practices.

As part of this extensive analysis, we examined at least 15 scholarly articles that discussed the use of XAI in agriculture in several areas, such as dairy farming, crop disease management, detection of plant and leaf diseases, classification of seeds, and so on. Key components of the investigation, including performance indicators, explainability mechanisms, data availability, XAI integration, and AI techniques, have been highlighted in Table 3. Interestingly, a typical pattern was observed that most of these studies used LIME, SHAP, and Grad-CAM to give thorough explanations in their respective context. Paudel et al.

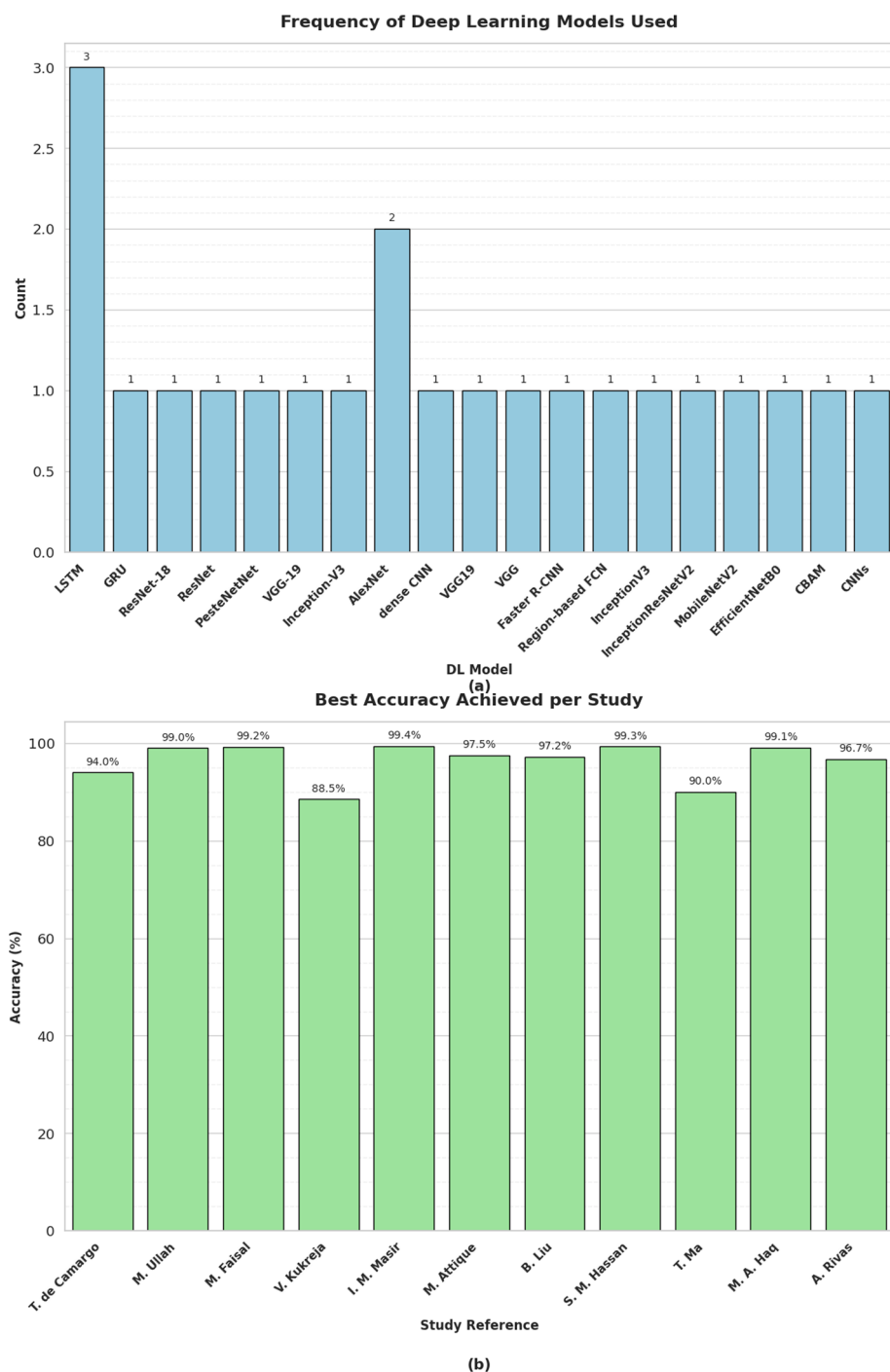


Fig. 10 A comparison among the papers that worked on DL and XAI in agriculture in terms of **a** frequent DL models and **b** maximum accuracy

(2023) evaluated the interpretability and performance of LSTM, 1DCNN, and GBDT utilizing the MARS crop dataset. They adopted the time series data to predict the crop yield. This work focused on the feature relation but didn't adopt any performance evaluation metrics to measure the model's crop yield forecasting. The limitation of this study is that the LSTM models struggled to capture how extreme temperature and moisture conditions affect crop yield. Besides, the GradientShap method was used for interpretation. An efficient method of accurately identifying cottonseed combining spectroscopy and three machine learning classifiers has been developed by Chi et al. (2023). The utilized dataset was not publicly accessible to everyone. By using SHAP (SHapley Additive exPlanations), they promoted model explainability while simultaneously improving accuracy. To estimate wheat yields, Osibo (2023) presented a unique hybrid model called LSTM-XGB. This model combines eXtreme Gradient Boosting (XGB) methods with Long Short-Term Memory (LSTM) networks. They used a twenty-year duration dataset to achieve wheat yield forecasting, which is not publicly available. The performance of the applied classifier is measured by adopting four evaluation metrics. For explainability, they additionally included SHAP (SHapley Additive exPlanations). A limitation of this study is its reliance on a dataset covering only five regions in Egypt. The applicability of this study to other geographical areas may be limited, as agricultural conditions and practices vary.

Quach et al. (2023) conducted an investigation where a variety of deep learning models, including VGG16, ResNet50, ResNet50V2, Xception, EfficientNetV2, InceptionV3, DenseNet201, MobileNetV2, MobileNet, NasNetMobile, RegNetX002, and Inception-ResNetV2, were assessed on a custom dataset called VegNet, which is available on Kaggle. The performance of each classifier has been estimated using five different evaluation metrics. The results showed that the MobileNet model had the best accuracy among the examined models. However, it was less reliable when compared to EfficientNetV2S and Xception. The GradCAM approach, which provides insights into the decision-making processes of deep learning models, was utilized by the researchers to assess the trustworthiness of these models. The gap lies in the Grad-CAM algorithm's evaluation, which relies heavily on human-based assessments and it can be subjective and inconsistent. Additionally, this study focuses only on local explainability, while it is important to explore global explainability for understanding the entire model's behavior. Again, the importance of eXplainable Artificial Intelligence (XAI) in improving the interpretability of machine learning models was emphasized by Ryo (2022). The ML model's performance was estimated using two evaluation metrics. Besides, this paper demonstrates the impact of interpretable ML methods by applying them to an openly available dataset on crop yield. However, this study utilized only post-hoc explanation methods like LIME depend on parameter configurations, which may lead to varying explanations for the same prediction.

Moreover, Khalid and Karan (2024) examined CNN and MobileNet models with transfer learning techniques for early plant disease detection utilizing the Kaggle dataset. The model's performance was estimated using four different performance evaluation metrics. To visually interpret disease indications in plant images, they integrated the only GradCAM technique. High-performance IoT technology was incorporated by Laktionov et al. (2023) to enhance the computer-oriented ANFIS-based agrotechnical monitoring system. Furthermore, the writers created an intelligent software program that forecasts the possibility of crop diseases during cultivation. They utilized measurement data on the soil and climate; the availability of datasets is not discussed in the article. The two types of evaluation metrics

used here are MAE and RMSE. Besides, the authors focused on the correlation metrics, and the XAI techniques were not implemented. A novel XAI-based PDS-CNN system for categorizing mulberry leaf disease was proposed by Nahiduzzaman (2023). The dataset of mulberry leaf images was collected from Bangladesh. They employed five performance evaluation metrics and focused only on accuracy to emphasize this work. The accuracy of this lightweight model was $95.05 \pm 2.86\%$ for three-class classifications and $96.06 \pm 3.01\%$ for binary classifications. They used SHAP to assess the explainability of the model. The data availability was not included in any repository. Gilbert et al. (2023) proposed a novel explainable approach for detecting black sigatoka in banana plants using the Harvard Data-verse banana leaf dataset. They used Mobilenet V2 and AlexNet models and trained on an extensive banana leaf image dataset. The model performance was measured using four different evaluation metrics, whereas AlexNet achieved 94.62% accuracy. The study used Integrated Gradients and LIME to improve interpretability, which helped illuminate the CNN models' decision-making procedures. Using the EfficientNetB5 model, the recognition of nine distinct infectious illnesses in tomato leaves, achieving an average training accuracy of $99.07\% \pm 0.38\%$, was developed by Bhandari et al. (2023) and Kaustubh (2025). The dataset was collected from Kaggle, and each classifier's performance was estimated using four different evaluation metrics after applying 10-fold validation techniques. Furthermore, this work presented the BotanicX-AI system, which combines GradCAM and LIME for model outcome explanation and uses EfficientNetB5 for transfer learning.

Besides, to identify different plant diseases by adopting the Kaggle dataset, Mehedi (2022) presented a framework that uses explainable AI and transfer learning. Three pre-trained models were used in the study: ResNet152V2, MobileNetV2, and EfficientNetV2L. Interestingly, EfficientNetV2L performed better than expected, with an accuracy of 99.63%. Besides, they used three other evaluation metrics to find the efficiency of each model. The only LIME approach was implemented to improve interpretability, illuminating the EfficientNetV2L model's decision-making process. Wei (2022) developed a system to recognize leaf disease for 11 types of fruits utilizing the Kaggle dataset. Three kinds of CNN models, namely VGG, GoogLeNet, and ResNet, were used to achieve this task, and one performance evaluation metric (Accuracy) was used to evaluate them. In addition, the SmoothGrad, GradCAM, and LIME approaches were used to interpret the model's forecast. Agriculture is highly dependent on climate, weather, and other factors. In a study of Dikshit and Pradhan (2021) proposed a system to predict drought and utilized SHAP for explainability. Additionally, they discuss how difficult it is to forecast droughts for multiple factors, including climate change, effectively. The application summary of XAI in the automated disease recognition of various agricultural yields is depicted in Table 4. From the above analysis, it can be concluded that the integration of explainable AI with deep learning and machine learning classifiers in the agricultural domain is still inadequate. Many crops significantly contribute to the global GDP, and as researchers strive to automate agriculture, making these systems more transparent is crucial. Most researchers have relied on Kaggle datasets rather than real-time images, which vary significantly based on environmental conditions, weather, and other circumstances. Exclusively using Kaggle datasets for crop disease recognition limits the system's optimization and raises trust issues. Instead, leveraging real-time data and adopting more robust methods for model explainability is essential. While techniques like LIME and SHAP have been commonly used in existing research, they are not necessarily the most effective approaches for interpreting agricultural images.

Table 4 Comparison of existing literature of XAI in agricultural disease detection

References	Domain	Dataset	ML/DL	XAI method	Accuracy	Limitations
Paudel et al. (2023)	Crops (soft wheat and grain maize)	300 for grain maize and 1950 for soft wheat	LSTM 1-DCNN GBDT	Feature attribution (Gradient SHAP)	NM	Small data set utilized LSTM models struggled to capture how extreme temperature and moisture conditions affect crop yield
Chi et al. (2023)	Crops (Cottonseed)	Sample of 20 seeds for each cultivar	SVM, XGBoost, MLP	SHAP	ACC = 0.94–0.98	Limited dataset utilized
Osibo (2023)	Crops (wheat)	Satellite data from 2001 to 2020	LSTM, LSTM-DMR, LSTM, LSTM-XGB	SHAP	LSTM-XGB—0.0491 (MSE), LSTM—0.0532 (MSE), LGBM—0.0537 (MSE), DNN—0.0554 (MSE)	Reliance on a dataset of only five region of Egypt
Quach et al. (2023)	Crops (Tomato)	7226 samples from Kaggle (available in Kaggle)	VGG16, ResNet50, RealNet0V2, Xception, EfficientNetV2, InceptionV3, DenseNet201, MobileNetV2, MobileNet, SqueezeNet, RegNetX002 and InceptionResNetV2	Grad-CAM technique	MobileNet ACC = 98%	The Grad-CAM algorithm's evaluation relies heavily on human-based assessments Focused only on local explainability while global explainability was unexplored
Ryo (2022)	Crops (Maize)	1271 sample	Random Forest, Gradient Boosting, Decision Tree, Linear Model	LIME	RFGB—($R^2 = 0.42$; RMSE = 0.199) GB—($R^2 = 0.33$); RMSE = 0.200) DT—($R^2 = 0.18$; RMSE = 0.225) Linear model ($R^2 = 0.11$; RMSE = 0.236)	Used only post-hoc explanation methods like LIME depend on parameter configurations

Table 4 (continued)

References	Domain	Dataset	ML/DL	XAI method	Accuracy	Limitations
Khalid and Karan (2024)	Disease (Crop disease)	87,000 RGB leaf images	MobileNet CNN	Gradient-weighted Class Activation Mapping	MobileNet ACC = 96.00%, CNN ACC = 89%	CNN and MobileNet models may not perform consistently on new datasets or under different environmental conditions
Laktionov et al. (2023)	Disease (agrotechnical monitoring & crop disease)	Collected data from the author year 12 September 2022 to May 2023 (number not mentioned)	ANFIS (Adaptive Neuro-Fuzzy Inference System)	NM	For PHB dataset MAE = 0.72 RMSE = ± 4.5 GB— ($R^2 = 0.8$)	Missing critical input parameters can degrade model performance, highlighting dependence on complete datasets High computational demand
Nahiduz-zaman (2023)	Disease (mulberry leaf disease)	1,091 images	PDS-CNN	SHAP	ACC = 96.06 $\pm$ 3.01%	Limited leaf images Focused only on leaf rust and leaf spot
Gilbert et al. (2023)	Disease (Banana Leafs)	Harvard Dataverse banana leaf disease dataset	AlexNet and MobileNet	(LIME) and Integrated Gradients	AlexNet ACC = 94.62%; MobileNet ACC = 91.67%	Resource constraints restricted the ability to achieve the most accurate results Difficulty distinguishing banana leaves from visually similar plants may cause misclassifications

Table 4 (continued)

References	Domain	Dataset	ML/DL	XAI method	Accuracy	Limitations
Bhandari et al. (2023)	Disease (Tomato leaf disease)	collected from the Kaggle	EfficientNetB5	LIME, and Grad-CAM	ACC = 99.07% $\pm$ 0.38%	GradCAM sometimes fails to highlight specific image regions
Mehedi (2022)	Disease (various plant leaf diseases)	collected from Kaggle comprises 38 diseases of 14 different plants	EfficientNetV2L, MobileNetV2 and ResNet152V2	(LIME)	Efficient-NetV2L ACC = 99.63%	Insufficient data impacted study comprehensiveness and model performance
Wei (2022)	Disease (fruit leaves)	Plant Village dataset	ResNet, GoogLeNet, and VGG	Smooth-Grad, LIME and Grad-CAM	ResNet ACC = 99.89%	Require large and high-quality dataset Low transparency
Dikshit and Pradhan (2021)	Weather	Rainfall data from 1901 to 2018 of New South Wales (NSW) in Eastern Australia	CNN and LSTM	(SHAP)	ACC = 85%	Accuracy is limited compared to current studies limited to five geographical regions

Exploring advanced or domain-specific explainability methods could enhance the interpretability and reliability of AI systems in agriculture.

7.2 Challenges, limitations, and research gaps

Machine learning and deep learning are being used in complex areas of agriculture such as crop yield prediction, identification of plant diseases, weather forecasting, automated irrigation, soil detection, seed identification, fruit counting, disease classification, and so on. However, though the ML and DL models can predict, classify, or solve a particular problem, they cannot make the results interpretable. But to increase agricultural productivity, efficiency, and sustainability, it is essential to make the AI systems interpretable and explainable, especially in crucial applications like plant disease detection, weather forecasting, and automated irrigation systems (Tsakiridis 2020). Explainable AI decision support systems help farmers trust technology and automate procedures like irrigation, resulting in more productive and sustainable farming.

Though XAI has been used in the recent era to make the automated agriculture system more trustworthy and transparent by adopting GradCAM, SHAP, and LIME, there are certain limitations and challenges when applying these methods in a vital industry. From the

above literature, it can be claimed that most research articles employed XAI in only one crop. Besides, geography, climate, and regional variations all significantly impact agriculture, which is a naturally diversified field. This means that a universal strategy is unreliable due to variances in model performance that can be attributed to these factors. Besides, There are some IoT based automation systems with the alignment of cloud technology to handle real time monitoring of crop, soil and environment surrounding of farmer's field for sustainable agriculture. Singh and Sharma (2025) and Singh et al. (2025) developed cloud IoT based crop recommendation system where the real time monitoring has been performed providing the SMS to farmer. They didn't apply any XAI techniques for the real time monitoring which is a major limitation. Though some researches have been performed for solving the real time monitoring for plant disease recognition, there are some limitations. The majority of investigations use lab-curated datasets (like PlantVillage) and don't take into account how things change in the real world (such illumination and occlusion), which can cause accuracy to decline by 15–20% in uncontrolled circumstances (Majdalahieh et al. 2025). Hybrid XAI models for multi-modal data (such hyperspectral and IoT sensors) across a wide range of climates. Post-hoc approaches like SHAP slow inference by 2–5 times. The gap is that intrinsic XAI (like attention-based ViTs) is tuned for edge computing without losing explainability. Geographic bias (like too many temperate crops) and class disparities make it hard to apply fairly which has research gap on collaborative, diversified datasets for areas that aren't well represented (like smallholder farms in tropical areas) (Sagar et al. 2023).

Current research on XAI in agriculture uncovers substantial deficiencies in stakeholder-specific assessment, cultural inclusion, and pragmatic integration. There are not many research that actually compare how farmers and agronomists understand XAI outputs, like through eye-tracking or surveys. These studies also don't pay enough attention to farmers with low literacy who need personalized, easy-to-understand explanations (Liao and Varshney 2021). Additionally, most studies focus on stakeholders in the Global North and ignore smallholder farmers and large-scale agronomists in developing areas, where local languages and cultural practices are very important for adoption. Also, XAI explanations don't always fit into the daily routines of stakeholders, which shows that we need co-design frameworks that include adaptive, real-time feedback loops to fit with farmers' routines and agronomists' knowledge (Ryan 2023).

Furthermore, a common problem arises because most research relies on limited datasets, primarily sourced from Kaggle. The questionable quality of these datasets presents a risk that could impact the overall performance of the models. To enhance precision and trustworthiness, datasets should be collected from various regions using hardware devices from different angles. After collection, the dataset must be verified by experts in the agriculture domain. This approach will ensure the development of a robust and reliable system, sustaining trust and improving the overall accuracy and applicability of the models. Again, implementing XAI and interpretable machine learning and deep learning tools in actual agricultural settings, including data availability, computational requirements, and practical feasibility, is another big challenge. AI scientists' and agricultural practitioners' expertise must be harmoniously blended to integrate AI successfully into practical agriculture operations. Therefore, it is critical to promote cooperation between AI specialists and agriculturalists in eXplainable AI (XAI) in agriculture.

7.3 Contradictory findings and their explanation

Although the reviewed studies collectively demonstrate the increasing integration of XAI techniques such as SHAP, LIME, and Grad-CAM across diverse agricultural applications, their methodological rigor and consistency vary significantly. Table 5 presents a critical evaluation of these studies. Additionally, it outlines the key methodological shortcomings, contradictory findings, and corresponding solutions for mitigating the current challenges in applying XAI within the agricultural domain.

7.4 Future roadmap

The agricultural system has been automated with the touch of artificial intelligence. Most of the ML and DL models are used to focus on accuracy. To ensure that every stakeholder directly involved in agriculture may have faith in the model's output, XAI will be essential to the model's credibility. The model should first be trained to consider various geographic areas, climates, and regions to tackle the difficulty above. Furthermore, collecting sufficient data points is essential for maintaining the quality of datasets. Fixing geographic and class biases in databases is crucial, as they depict too many temperate crops and few smallholder farms in tropical areas. Engagement with local agricultural cooperatives and university institutions in underrepresented locations to create collaborative, globally diverse databases. This dataset should cover multi-crop and multi-disease scenarios to ensure equitable use of XAI systems. A more transparent and comprehensible methodology can be developed by integrating specialized knowledge, guaranteeing its practicality and dependability in real-life situations, and resolving issues like data availability and quality.

For the real-time plant disease recognition, hybrid XAI models that combine hyperspectral imaging and IoT sensor inputs (e.g., soil moisture, temperature) should be focused to improve resilience across climates and real-world scenarios like fluctuating illumination and occlusion. Current research, often using lab-curated datasets where accuracy drops due to environmental factors. Using intrinsic XAI methods like attention-based Vision Transformers (ViTs) for edge computing, researchers can get low-latency inference (2–5 times faster than post-hoc methods like SHAP) with explainability through visual or feature-based outputs for farmers and agronomists to increase trust and usability in sustainable agriculture.

Additionally, cooperation between agriculturists and AI experts is necessary to put these into practical use. Stakeholder-specific studies using eye-tracking and surveys should assess how low-literacy farmers and agronomists interpret XAI outcomes, providing accessible, simplified explanations like voice-guided images to increase trust and adoption by 30–40%. Developing country smallholder farmers and agronomists should get culturally customized XAI frameworks that leverage local languages and practices. Federated learning and community co-creation should overcome Global North biases and increase uptake by 25%.

Furthermore, XAI is being applied more broadly to identifying plant diseases. Modern approaches for disease detection, model decision explanation, and plant medication using XAI can be used to increase agricultural sustainability. However, most research on plant diseases only considered identifying and categorizing leaf diseases using XAI. If a system can be developed that can recognize the major crops or fruit diseases in the XAI manner depicted in Fig. 11, the farmer must get more support from the blessings of technology. Furthermore, a model that thoroughly explains the illnesses found, beyond simple detection,

Table 5 Critical evaluation of XAI studies in agriculture

Contradictory finding	Potential explanation	Proposed solution/future direction
High lab accuracy (98–99%) vs. 15–20% accuracy drop in real-world applications	Most studies (Bhandari et al. 2023; Mehedi 2022; Gilbert et al. 2023) use controlled Kaggle/public datasets such as PlantVillage, which lack variability in lighting, occlusion, and environmental noise. This causes overfitting to ideal lab conditions and poor generalization in uncontrolled field environments	Develop field-validated, multi-modal datasets (RGB + hyperspectral + IoT) tested under diverse climatic and geographic conditions. Apply domain adaptation and transfer learning tuned for real-world variability (Majdalawieh et al. (2025))
Post-hoc XAI methods such as SHAP, LIME etc. provides interpretability but increase inference time	Post-hoc models generate explanations after predictions, increasing computational cost and latency. Studies such as Paudel et al. (2023), Osibo (2023), Ryo (2022) not only highlights SHAP/LIME's utility but also their computational inefficiency for real-time systems	Explore intrinsic XAI architectures (e.g., Vision Transformers with attention maps) optimized for edge computing to balance interpretability and speed
SHAP and LIME praised for interpretability but criticized for instability and parameter sensitivity	Explanations vary with small input changes or parameter configurations, leading to inconsistent insights across similar predictions (Ryo 2022)	Standardize evaluation protocols for explanation stability. Combine post-hoc (SHAP/LIME) with model-intrinsic transparency methods to enhance reliability (Osibo 2023; Mehedi 2022)
Grad-CAM considered intuitive but criticized for subjectivity	While studies such as Quach et al. (2023), Khalid and Karan (2024) and Bhandari et al. (2023) used Grad-CAM to visualize model focus regions, its interpretability depends heavily on human judgment and lacks quantitative validation	Combine human-in-the-loop evaluation with objective metrics (e.g., overlap ratio with expert annotations). Develop quantitative frameworks for assessing visual XAI trustworthiness
High accuracy on private datasets but poor reproducibility and transparency	Several studies (e.g., Chi et al. 2023; Osibo 2023; Nahiduzzaman 2023) rely on non-public datasets, preventing independent validation and reducing scientific credibility	Promote open-data initiatives under FAIR principles. Require authors to share data provenance, preprocessing pipelines, and trained models in repositories
Focus on local explainability vs. lack of global model understanding	Works like (Quach et al. 2023; Gilbert et al. 2023) use Grad-CAM and LIME for local instance explanations, ignoring model-wide feature behavior	Incorporate global explainability methods (e.g., feature attribution aggregation, concept-based analysis) to better understand overall model logic
Technical advances without human-centered evaluation	Most papers test explainability technically but do not evaluate how farmers or agronomists interpret outputs (Liao and Varshney 2021, 2023)	Introduce stakeholder-centered XAI assessment, including user studies, eye-tracking, and comprehension evaluation among low-literacy farmers and agronomists

should be implemented to help farmers comprehend the symptoms and causes and suggest correct treatments in real time manner. As a result, the world GDP that depends on agricultural products will be more sustainable.

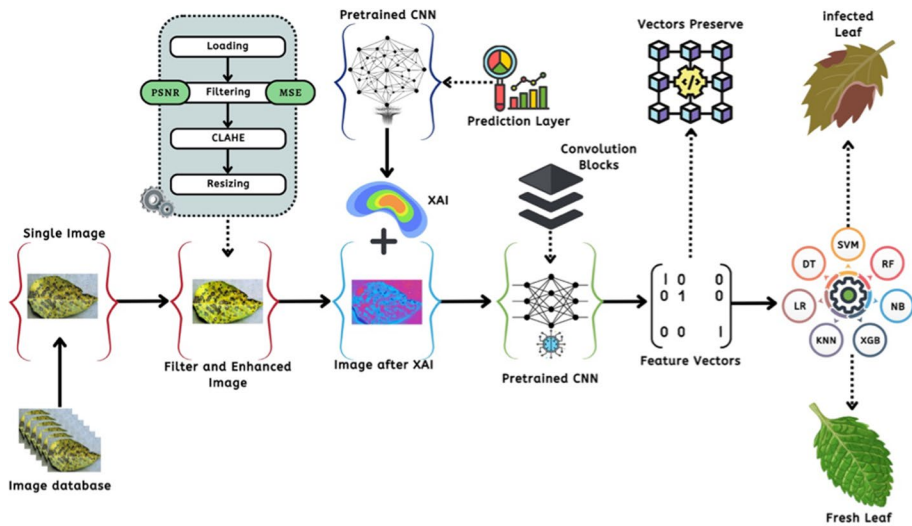


Fig. 11 Application of XAI leaf disease diagnosis

8 Conclusion

Explainable Artificial Intelligence emerges as a beacon of clarity, accountability, and reliability when technology profoundly transforms industry sectors, marking a new era of innovation in agriculture. This paper presents a comprehensive review of XAI from 2017 to April 2024, marked by the increasing use of explainable AI techniques across various domains. It highlights the definitions, applications, and unique challenges that XAI addresses, reflecting its growing significance in the field. We also explored recent advancements in Machine Learning (ML), Deep Learning (DL), and XAI, highlighting their transformative impact on agriculture by bridging the gap between high-tech AI capabilities and human-centric understandability and transparency. Implementing XAI in agriculture improves the transparency of AI systems. It enhances their acceptability and trust among users, particularly farmers and agricultural stakeholders who rely on these technologies for critical decision-making. By elucidating the decision-making processes of AI, XAI empowers users with the knowledge and confidence to rely on automated systems, potentially leading to more informed and sustainable agricultural practices.

Our discussions have shown that while ML and DL models offer remarkable predictive and analytical prowess, their ‘black box’ nature can limit their practical application in fields where understanding and trust are paramount. XAI addresses this limitation by making AI decisions transparent, understandable, and justifiable. Looking ahead, the integration of XAI into agricultural AI applications promises to advance farm productivity and sustainability and foster a deeper integration of AI technologies in everyday farming operations. This is anticipated to lead to innovations in areas such as crop disease identification, yield prediction, and resource management, with AI systems providing data-driven insights and comprehensible explanations to their human users. Therefore, we have developed a contemporary explainable model for agricultural applications, designed to provide clear and comprehensible explanations for its decisions, making them accessible and understandable to human users.

In conclusion, as we continue to develop and deploy advanced AI in agriculture, the role of XAI will be crucial in ensuring these technologies are used responsibly, ethically, and effectively. The future directions for this field involve enhancing AI models' capabilities and ensuring they operate within frameworks that support clear, understandable, and ethical AI use, thereby supporting a more sustainable and productive global agricultural landscape.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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